

Signals Lab 2 - Impulse Response, Step Response, and RC Circuits

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Introduction

Our objectives for this lab were to examine how a short input pulse can be used to approximate the impulse response for a system $h(t)$, determine how short of a pulse can be used for this approximation, and test whether, as $T \rightarrow \infty$ the response of a system depends on the shape of the pulse. We aimed to accomplish these objectives through hand analysis of the pulse and step responses of the two-stage RC cascade shown in Figure 1, numerical modeling of the two-stage RC circuit with python, and experimental construction and measurement of the two-stage RC circuit on a breadboard.

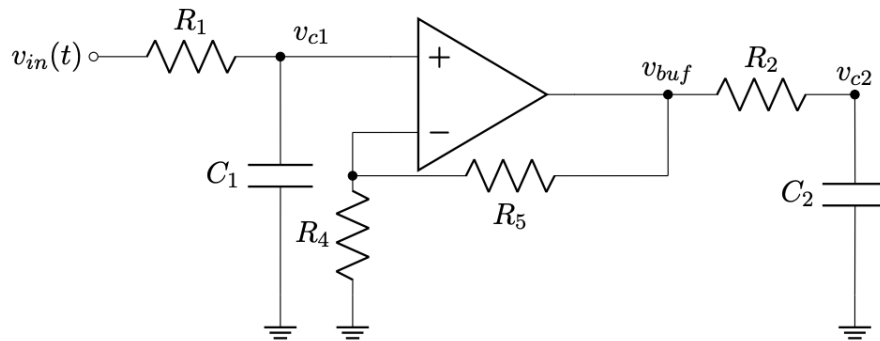


Figure 1: Two-stage RC Cascade with an input RC circuit, non-inverting amplifier circuit, and output RC circuit

Task 1: Prelab Derivations

1. Time constants and gain

$$\begin{aligned}\tau_1 &= R_1 C_1 = (10 \text{ k}\Omega)(1 \text{ }\mu\text{F}) = 1.0 \times 10^{-2} \text{ s} \\ \tau_2 &= R_2 C_2 = (4.7 \text{ k}\Omega)(1 \text{ }\mu\text{F}) = 4.7 \times 10^{-3} \text{ s} \\ G &= 1 + \frac{R_5}{R_4} = 1 + \frac{4.7 \text{ k}\Omega}{1 \text{ k}\Omega} = 5.7\end{aligned}$$

2. Second-order ODE for $v_{c2}(t)$

The two stages are described by

$$\begin{aligned}\dot{v}_{c1}(t) &= \frac{1}{\tau_1}(v_{in}(t) - v_{c1}(t)), \\ \dot{v}_{c2}(t) &= \frac{1}{\tau_2}(G v_{c1}(t) - v_{c2}(t)).\end{aligned}$$

Differentiating the second equation and substituting $\dot{v}_{c1}$ from the first:

$$\begin{aligned}\ddot{v}_{c2}(t) &= \frac{G}{\tau_2}\dot{v}_{c1}(t) - \frac{1}{\tau_2}\dot{v}_{c2}(t) \\ &= \frac{G}{\tau_1\tau_2}(v_{in}(t) - v_{c1}(t)) - \frac{1}{\tau_2}\dot{v}_{c2}(t).\end{aligned}$$

Solving the second stage equation for v_{c1} gives

$$v_{c1}(t) = \frac{\tau_2 \dot{v}_{c2}(t) + v_{c2}(t)}{G},$$

and substituting this in:

$$\begin{aligned}\ddot{v}_{c2}(t) &= \frac{G}{\tau_1\tau_2}v_{in}(t) - \frac{1}{\tau_1\tau_2}(\tau_2\dot{v}_{c2}(t) + v_{c2}(t)) - \frac{1}{\tau_2}\dot{v}_{c2}(t) \\ &= \frac{G}{\tau_1\tau_2}v_{in}(t) - \frac{1}{\tau_1}\dot{v}_{c2}(t) - \frac{1}{\tau_1\tau_2}v_{c2}(t) - \frac{1}{\tau_2}\dot{v}_{c2}(t).\end{aligned}$$

Collecting terms:

$$\boxed{\ddot{v}_{c2}(t) + \left(\frac{1}{\tau_1} + \frac{1}{\tau_2}\right)\dot{v}_{c2}(t) + \frac{1}{\tau_1\tau_2}v_{c2}(t) = \frac{G}{\tau_1\tau_2}v_{in}(t)}$$

3. Impulse response $h(t)$

For an impulse input, the first stage output is the single-stage RC impulse response (Section 1.1 with $\tau = \tau_1$):

$$v_{c1}(t) = \frac{1}{\tau_1}e^{-t/\tau_1}u(t).$$

Plugging this into the second stage ODE, for $t \geq 0$:

$$\dot{v}_{c2}(t) + \frac{1}{\tau_2}v_{c2}(t) = \frac{G}{\tau_1\tau_2}e^{-t/\tau_1}.$$

Using the integrating factor $\mu = e^{t/\tau_2}$:

$$v_{c2}(t) = e^{-t/\tau_2} \left(\int \frac{G}{\tau_1 \tau_2} e^{t/\tau_2 - t/\tau_1} dt + C \right).$$

The exponent simplifies to

$$\frac{t}{\tau_2} - \frac{t}{\tau_1} = \frac{t(\tau_1 - \tau_2)}{\tau_1 \tau_2},$$

so the integral is

$$\int \frac{G}{\tau_1 \tau_2} e^{\frac{t(\tau_1 - \tau_2)}{\tau_1 \tau_2}} dt = \frac{G}{\tau_1 - \tau_2} e^{\frac{t(\tau_1 - \tau_2)}{\tau_1 \tau_2}},$$

which gives

$$v_{c2}(t) = \frac{G}{\tau_1 - \tau_2} e^{-t/\tau_1} + C e^{-t/\tau_2}.$$

The initial condition $v_{c2}(0) = 0$ gives $C = -\frac{G}{\tau_1 - \tau_2}$. Therefore

$$h(t) = \frac{G}{\tau_1 - \tau_2} (e^{-t/\tau_1} - e^{-t/\tau_2}) u(t)$$

Figure 2 shows $h(t)$ for the given component values.

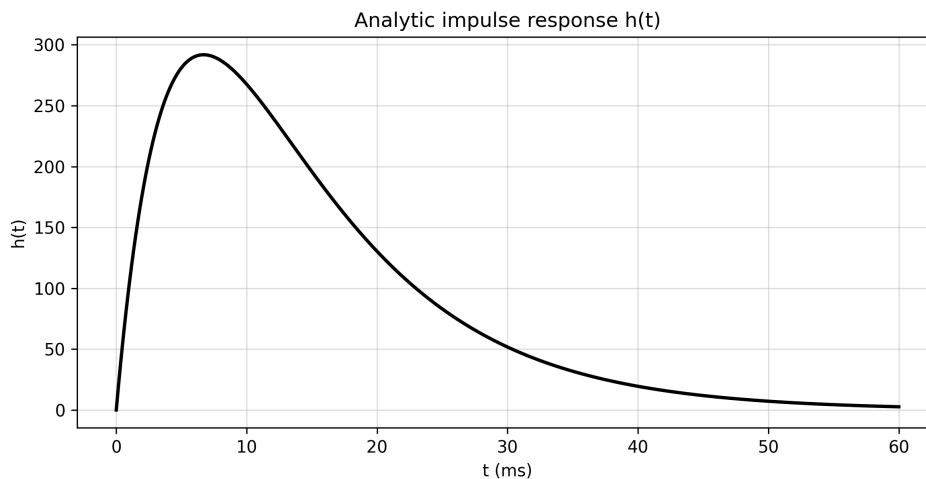


Figure 2: Analytic impulse response $h(t)$.

4. Step response $s(t)$

$$\begin{aligned} s(t) &= \int_0^t h(\sigma) d\sigma = \frac{G}{\tau_1 - \tau_2} \int_0^t (e^{-\sigma/\tau_1} - e^{-\sigma/\tau_2}) d\sigma \\ &= \frac{G}{\tau_1 - \tau_2} \left[-\tau_1 e^{-\sigma/\tau_1} + \tau_2 e^{-\sigma/\tau_2} \right]_{\sigma=0}^{\sigma=t} \\ &= \frac{G}{\tau_1 - \tau_2} \left[-\tau_1 e^{-t/\tau_1} + \tau_2 e^{-t/\tau_2} + \tau_1 - \tau_2 \right] \end{aligned}$$

$$s(t) = \frac{G}{\tau_1 - \tau_2} \left[\tau_1 (1 - e^{-t/\tau_1}) + \tau_2 (e^{-t/\tau_2} - 1) \right]$$

5. DC gain

As $t \rightarrow \infty$ both exponentials go to zero:

$$\lim_{t \rightarrow \infty} s(t) = \frac{G}{\tau_1 - \tau_2} [\tau_1 - \tau_2] = G = 1 + \frac{R_5}{R_4}.$$

6. Pulse response by superposition

A rectangular pulse of height A and width T is the difference of two shifted steps:

$$v_{in}(t) = A u(t) - A u(t - T) = \begin{cases} A, & 0 \leq t < T \\ 0, & \text{otherwise.} \end{cases}$$

The circuit is linear (its ODEs are linear) and time-invariant (the coefficients are constant), so scaling an input scales the output and delaying an input delays the output by the same amount. The response to a unit step is $s(t)$, so

$$A u(t) - A u(t - T) \xrightarrow{\text{LTI system}} A s(t) - A s(t - T),$$

and therefore

$$\boxed{v_{out}(t) = A[s(t) - s(t - T)]}$$

7. $v_{out}(t)$ using $s(t)$

For $0 \leq t < T$, $s(t - T) = 0$, so $v_{out}(t) = A s(t)$. For $t \geq T$:

$$\begin{aligned} v_{out}(t) &= A[s(t) - s(t - T)] \\ &= \frac{AG}{\tau_1 - \tau_2} \left([\tau_1(1 - e^{-t/\tau_1}) + \tau_2(e^{-t/\tau_2} - 1)] \right. \\ &\quad \left. - [\tau_1(1 - e^{-(t-T)/\tau_1}) + \tau_2(e^{-(t-T)/\tau_2} - 1)] \right) \\ &= \frac{AG}{\tau_1 - \tau_2} \left[\tau_1 (e^{-(t-T)/\tau_1} - e^{-t/\tau_1}) + \tau_2 (e^{-t/\tau_2} - e^{-(t-T)/\tau_2}) \right] \end{aligned}$$

8. Maximum T for 1% error

From Section 1.2, a 1% error requires $T \lesssim 0.02 \tau$, where τ is the smallest time constant of the system. Here the smallest time constant is $\tau_2 = 4.7 \times 10^{-3}$ s, so

$$T \lesssim 0.02 (4.7 \times 10^{-3} \text{ s}) = 9.4 \times 10^{-5} \text{ s} = 94 \text{ } \mu\text{s}.$$

Task 2.2: Rectangular Pulses

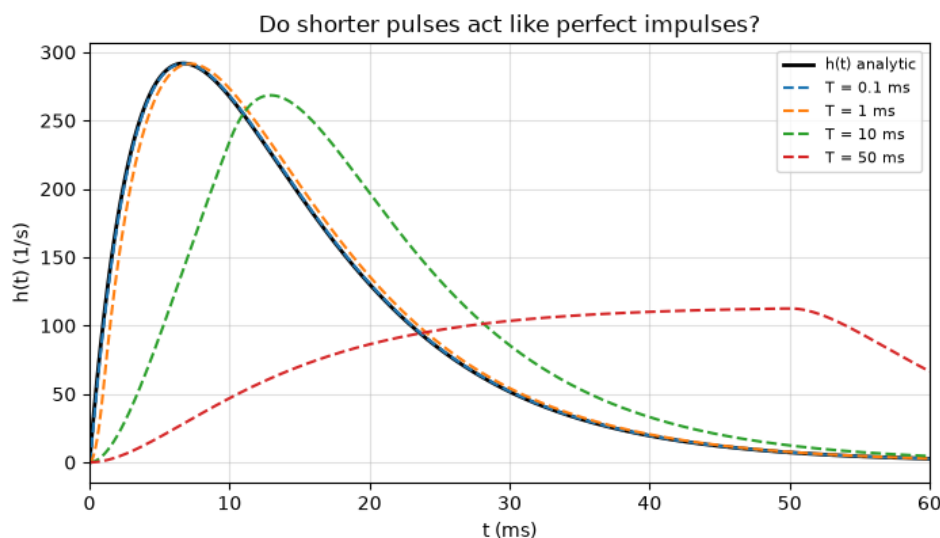


Figure 3: Four pulse widths compared to the analytic $h(t)$

Which T values collapse onto $h(t)$

Figure 3 shows the output divided by the pulse area ($A \cdot T$) for four different pulse widths, plotted against the analytic $h(t)$. The shortest pulse, $T = 0.1$ ms, collapsed almost perfectly onto $h(t)$, while the second shortest pulse, $T = 1$ ms, stays quite close to $h(t)$ and has the same overall shape, making it also work as a decent approximation of the impulse response. The 10 ms and 50 ms pulses do not collapse onto $h(t)$ at all. The 10 ms curve peaks lower and later than $h(t)$, and the 50 ms curve just slowly rises to a flat spot around 110 before falling off, which looks a lot more like a step response than an impulse response.

This lines up with the rule $T \lesssim 0.02 \tau_{min}$. The smallest time constant is $\tau_2 = 4.7$ ms, so the pulse needs to be shorter than $0.02(4.7 \text{ ms}) \approx 94 \mu\text{s}$ to be within 1% of $h(t)$. The 0.1 ms pulse is right at that limit, so it matches best. The 1 ms pulse is about 10 times over the limit, so it is close but a little off (roughly 5 to 10% error). The 10 ms and 50 ms pulses are far over the limit, so they are nowhere near $h(t)$. Basically, the pulse has to be much shorter than the circuit's time constants. If it is not, the circuit has time to respond while the pulse is still on, and the output is no longer just the impulse response.

Effect of amplitude ($A = 5 \text{ V}$ vs. $A = 1 \text{ V}$, $T = 100 \text{ ms}$)

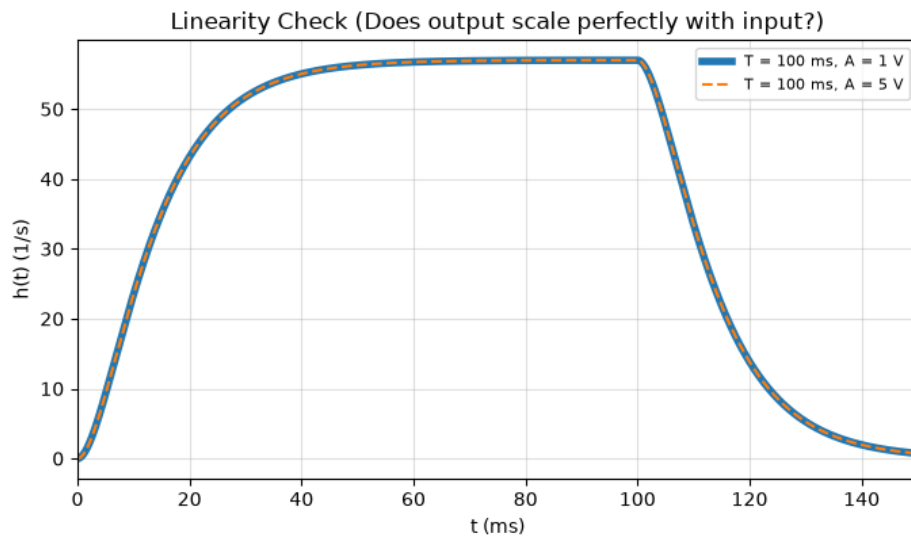


Figure 4: Area-normalized response for $A = 5 \text{ V}$ and $A = 1 \text{ V}$ at $T = 100 \text{ ms}$.

In Figure 4, the $A = 1 \text{ V}$ and $A = 5 \text{ V}$ curves land right on top of each other so the area-normalized curves are identical (the only difference is tiny numerical error from the solver). This shows that the system is linear. When the input gets 5 times bigger, the output also gets exactly 5 times bigger, so dividing by $A \cdot T$ cancels the amplitude out and same curve both times. The flat part of the curves sits at about 57, which is $G/T = 5.7/0.1 \text{ s}$. This makes sense because the output settles at the DC gain G times the input before the pulse ends. Neither curve matches $h(t)$ because 100 ms is way too long to act like an impulse, but that is fine since this test is only checking linearity.

Task 2.3: Triangular Pulse

For this task, the rectangular input was swapped for a triangular pulse with $T = 100 \mu\text{s}$ and a peak of $A = 2 \text{ V}$. The parts of the code that changed are below: the input array for plotting, the input inside `f_rhs` so the solver sees the triangle, and the normalization by the triangle's area.

```

1  A = 2.0          # peak of triangle (V)
2  T = 100e-6       # pulse width (s)
3
4  # triangular input array (for plotting)
5  vin = np.zeros(len(t_eval))
6  for i, t in enumerate(t_eval):
7      if 0 <= t <= T/2:
8          vin[i] = (2 * A / T) * t
9      elif T/2 < t <= T:
10         vin[i] = (2 * A / T) * (T - t)
11
12 # triangular input inside f_rhs so the solver sees it
13 def f_rhs(t, x):
14     v1, v2 = x
15     if 0 <= t <= T/2:
16         _vin = (2 * A / T) * t
17     elif T/2 < t <= T:
18         _vin = (2 * A / T) * (T - t)
19     else:
20         _vin = 0.0
21     return [(_vin - v1)/tau1, (G*v1 - v2)/tau2]
22
23 # normalize by triangle area = 0.5*A*T
24 area = 0.5 * A * T
25 axn.plot(t_eval * 1e3, v2 / area, "--", color="orange", label="
    Normalized Triangle")

```

Listing 1: Modified code for triangular input

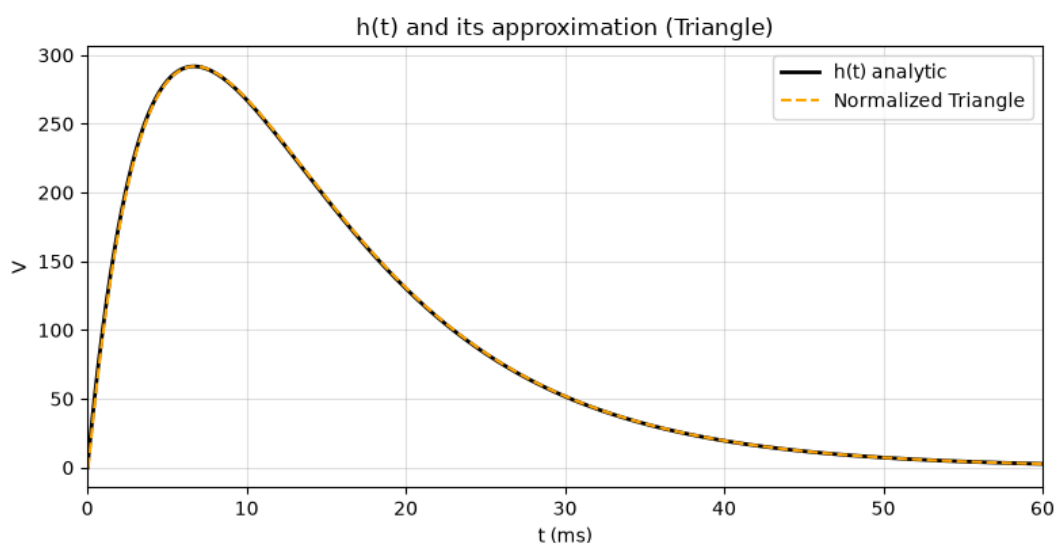


Figure 5: Triangular input and $v_{c2}/(0.5AT)$ overlaid with $h(t)$.

The triangle has an area of $\frac{1}{2}AT = \frac{1}{2}(2 \text{ V})(100 \mu\text{s}) = 1 \times 10^{-4} \text{ V}\cdot\text{s}$, which is the same as the 1 V, 100 μs rectangle from Task 2.2. Figure 5 shows the output divided by that area, and it lands right on top of $h(t)$. The peak is about 292 at around 6.7 ms, same as $h(t)$, and it stays within about 0.5% of $h(t)$ for the rest of the curve. So no, the shape of the pulse does not matter as long as the area is the same (and the pulse is short enough).

The math behind this is that the output is the input convolved with $h(t)$:

$$v_{c2}(t) = \int_0^T h(t - \tau) v_{in}(\tau) d\tau.$$

If T is really short compared to the time constants, then $h(t - \tau)$ barely changes while the pulse is on, so $h(t - \tau) \approx h(t)$ for $0 \leq \tau \leq T$. That lets $h(t)$ be pulled out of the integral:

$$v_{c2}(t) \approx h(t) \int_0^T v_{in}(\tau) d\tau = h(t) \cdot (\text{area of the pulse}).$$

So the only thing about the input that shows up in the output is its area, not its shape. That is why dividing by the area gives $h(t)$ for both the rectangle (AT) and the triangle ($\frac{1}{2}AT$). The tiny difference that is left comes from the pulse not being infinitely short.

Task 2.4: Step Response

For this task, the pulse was replaced with a step input of height $A = 1$ V that stays on for all $t \geq 0$. The modified parts of the code are below: the step input, the step inside `f_rhs` so the solver sees it, and the derivative of the step response using `np.gradient()`.

```

1  A = 1.0
2  t_end = 8 * tau_max           # no pulse now, so no T added
3  t_eval = np.linspace(0.0, t_end, 4000)
4  max_step = tau_min / 20.0
5
6  # step input is A for all t >= 0
7  vin = np.ones(len(t_eval)) * A
8
9  # f_rhs with a constant step input
10 def f_rhs(t, x):
11     v1, v2 = x
12     _vin = A                     # input is always A
13     return [(_vin - v1)/tau1, (G*v1 - v2)/tau2]
14
15 sol = solve_ivp(f_rhs, (0.0, t_end), [0.0, 0.0], t_eval=t_eval, rtol
16                 =1e-8, atol=1e-8, max_step=max_step)
17 v2_step = sol.y[1]
18
19 # differentiate the step response
20 ds_dt = np.gradient(v2_step, t_eval)

```

Listing 2: Modified code for step input and differentiation

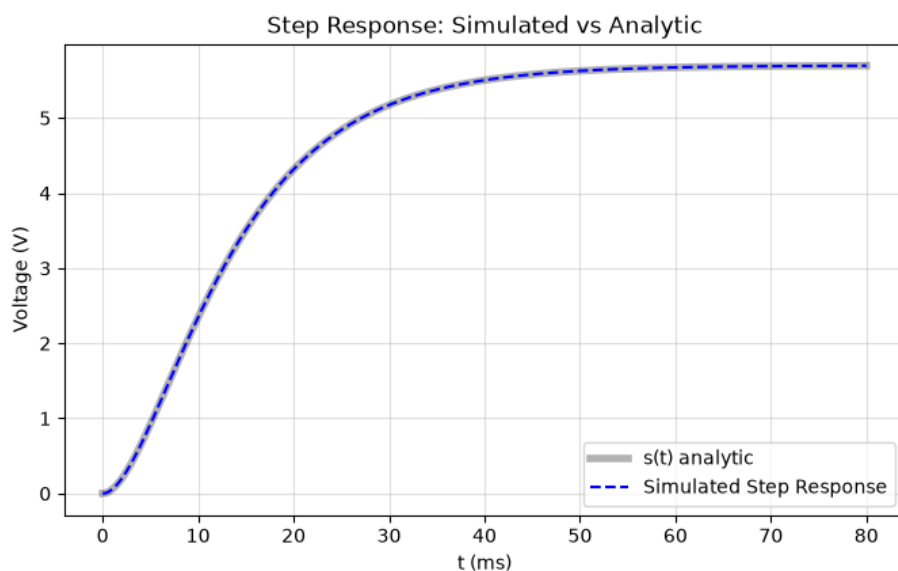


Figure 6: Simulated step response with analytic $s(t)$ overlaid.

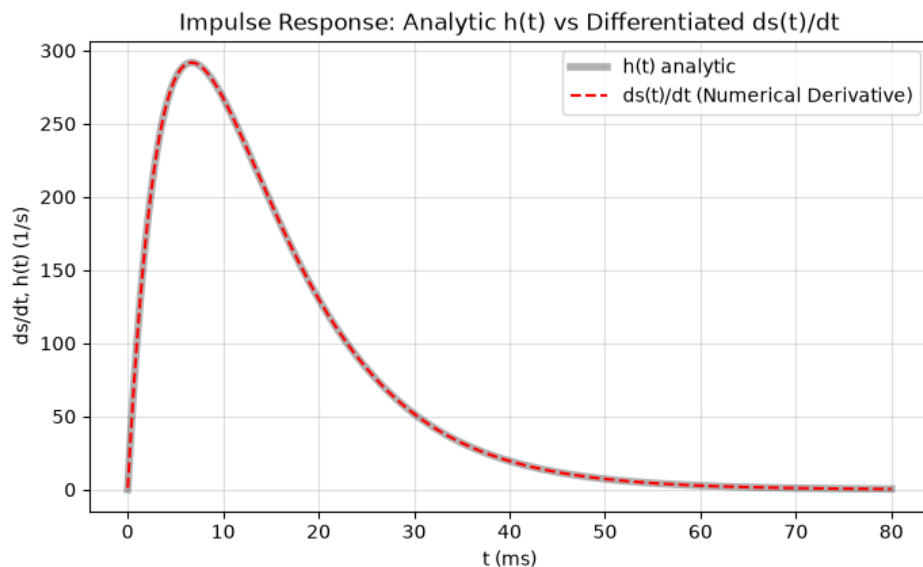


Figure 7: $h(t)$ overlaid with $ds(t)/dt$ computed numerically.

Figure 6 shows the simulated step response with the analytic $s(t)$ from Task 1 behind it. The two curves sit right on top of each other, so the simulation matches the math.

Are $h(t)$ and $ds(t)/dt$ the same?

Yes. In Figure 7, the numerical derivative of the step response lands right on top of the analytic $h(t)$. This makes sense because the step response is the integral of the impulse response,

$$s(t) = \int_0^t h(\sigma) d\sigma,$$

so taking the derivative of $s(t)$ just gives $h(t)$ back:

$$\frac{ds(t)}{dt} = h(t).$$

Final value and settling time

The step response levels off at about 5.696 V by $t = 80$ ms, which is $G = 5.7$ to within 0.1%. The tiny gap is just the last bit of the slow exponential not being gone yet. This matches the DC gain from Task 1, $G = 1 + R_5/R_4 = 5.7$, so a 1 V step ends up as 5.7 V at the output.

The settling time is mostly controlled by the slower time constant, $\tau_1 = 10$ ms. Writing $s(t)$ as

$$s(t) = G - \frac{G}{\tau_1 - \tau_2} (\tau_1 e^{-t/\tau_1} - \tau_2 e^{-t/\tau_2}),$$

the e^{-t/τ_2} term dies off faster, so at later times the gap to G is mostly just the e^{-t/τ_1} term. From the simulation, the output gets within 2% of G at about 45 ms and within 1% at about 52 ms. For a single RC circuit, 2% settling would be about $4\tau_1 = 40$ ms, so the second stage makes it take a little longer. The faster time constant τ_2 does not really set the settling time. It mostly shows up at the start, where the response begins flat and then curves upward instead of jumping up like a single RC circuit.

Task 3.1: Build and Instrument

The RC opamp circuit was built before any measurements were taken and everything was looked over by the group, the first build of the circuit was not working properly, due to a swapped resistor making the gain not boost to 5.7. Once that was resolved the opamp was also replaced just to be safe and the output was amplified as expected.

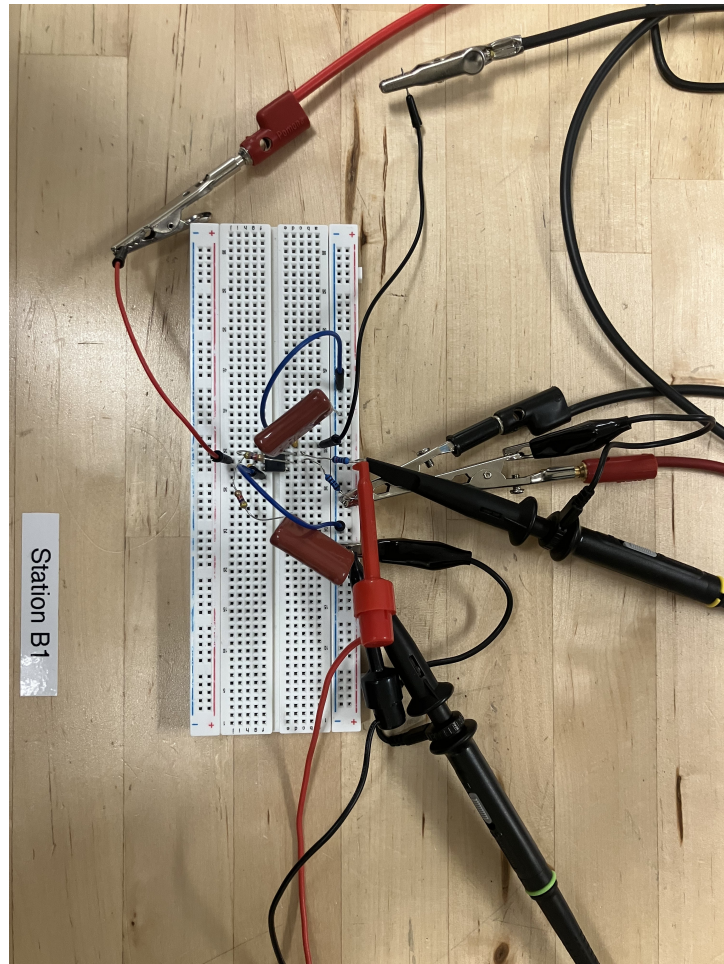


Figure 8: RC Opamp Circuit on Breadboard.

Task 3.2: Impulse Response Measurements

5 V, 100 μ s, 10 Hz

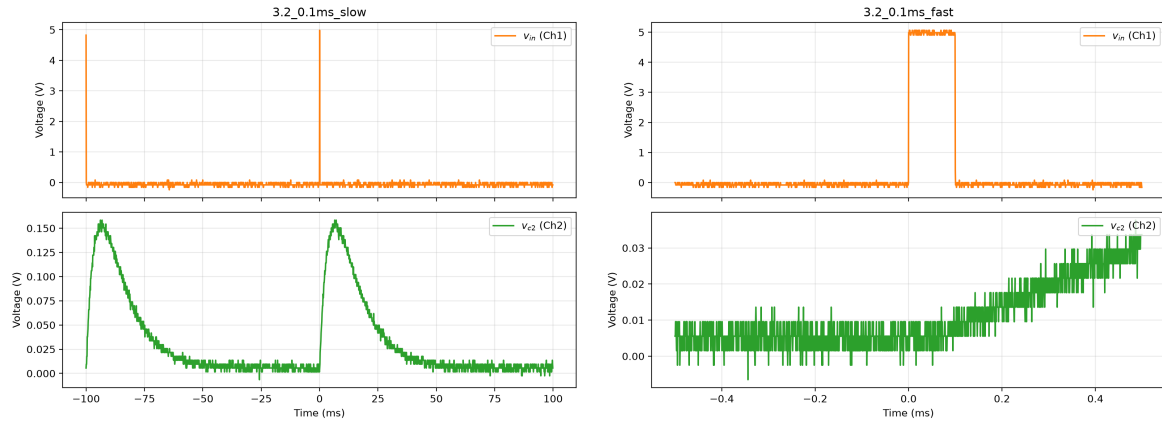


Figure 9: Slow and fast sweeps for the 100 μ s pulse.

5 V, 1 ms, 10 Hz

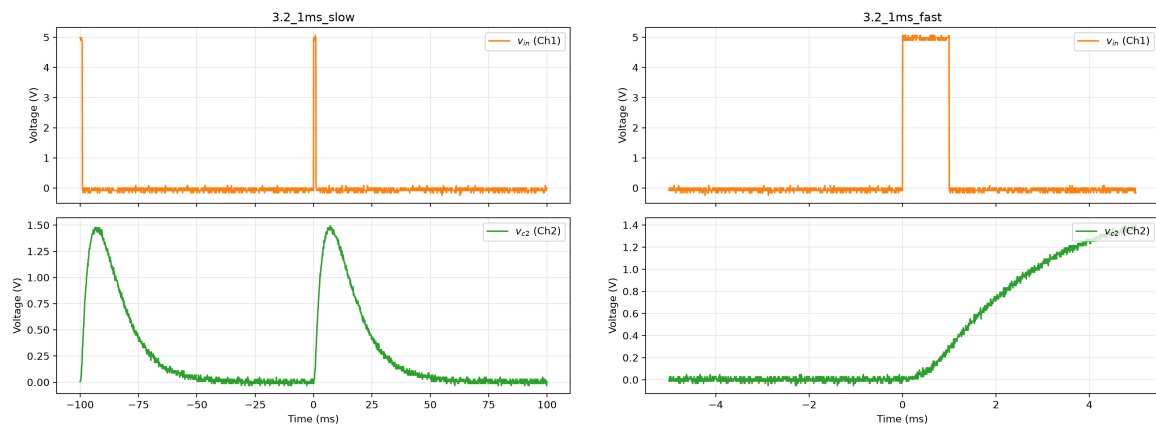


Figure 10: Slow and fast sweeps for the 1 ms pulse.

2 V, 10 ms, 10 Hz

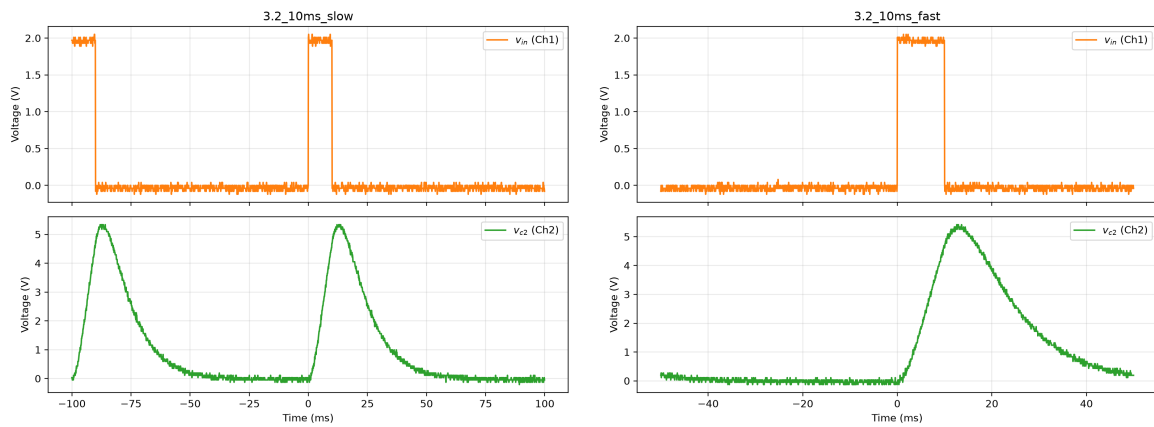


Figure 11: Slow and fast sweeps for the 10 ms pulse.

2 V, 50 ms, 1 Hz

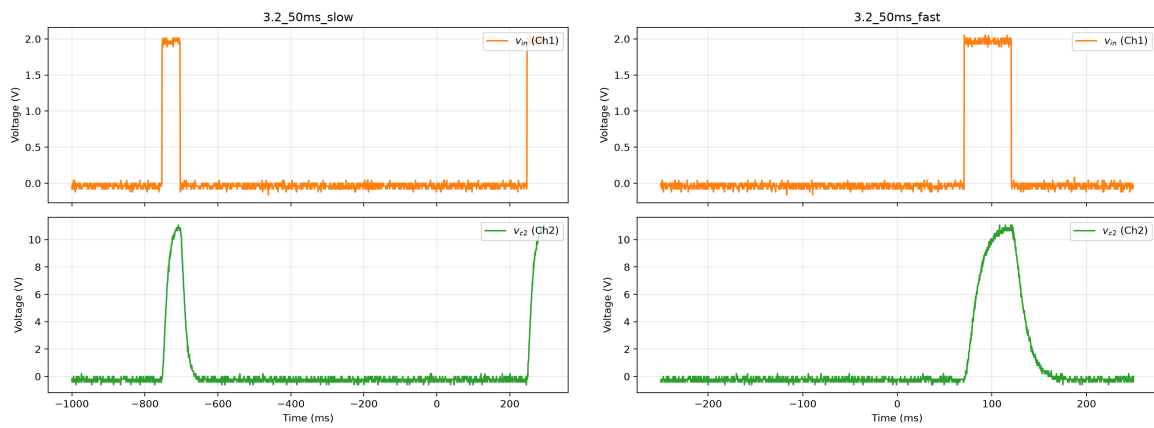


Figure 12: Slow and fast sweeps for the 50 ms pulse.

Normalized traces vs. $h(t)$

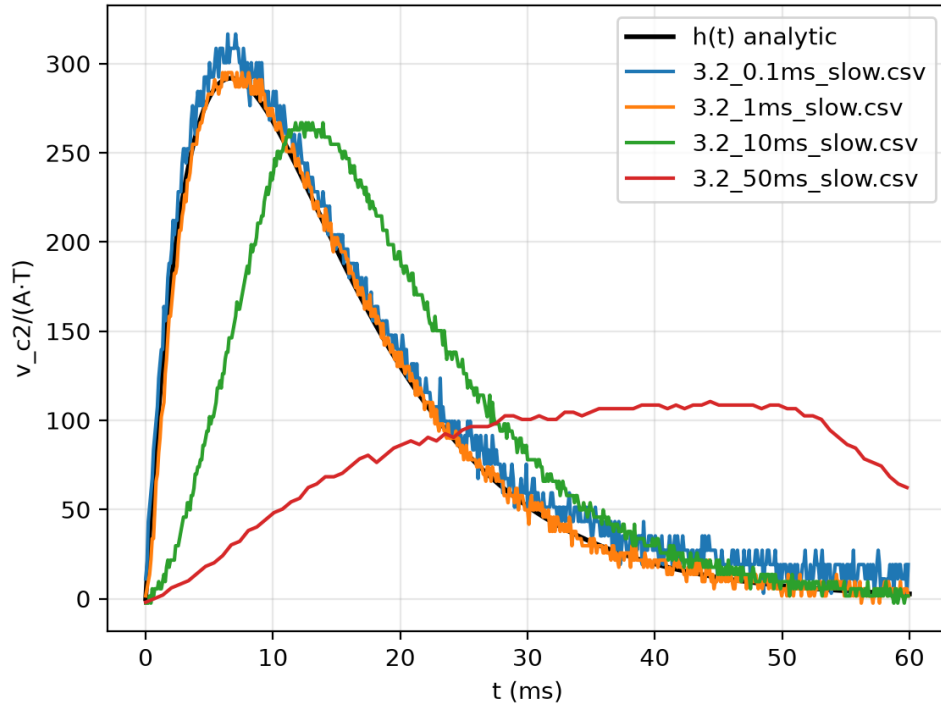


Figure 13: Four measured $v_{c2}(t)$ traces normalized by pulse area, with analytic $h(t)$.

Agreement with $h(t)$: The 100us and the 1ms have the best agreement with $h(t)$, the peaks and timing of the other graphs don't line up nearly as well and this is due to the signal not acting like an impulse at the higher widths, so the circuit has time to respond. Errors on top of this point to component tolerances, oscilloscope measurements, and a non-ideal pulse from the function generator.

Task 3.3: Step Response Measurements

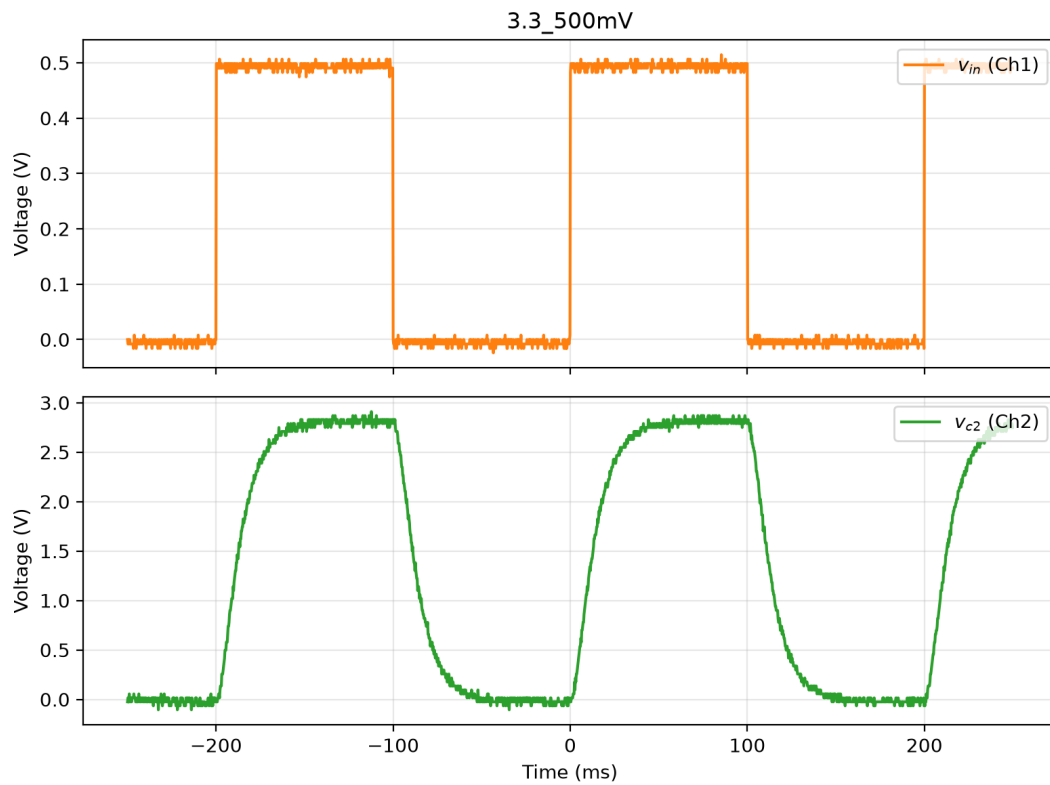


Figure 14: Measured response to a 500 mV step (same rise and fall time constants).

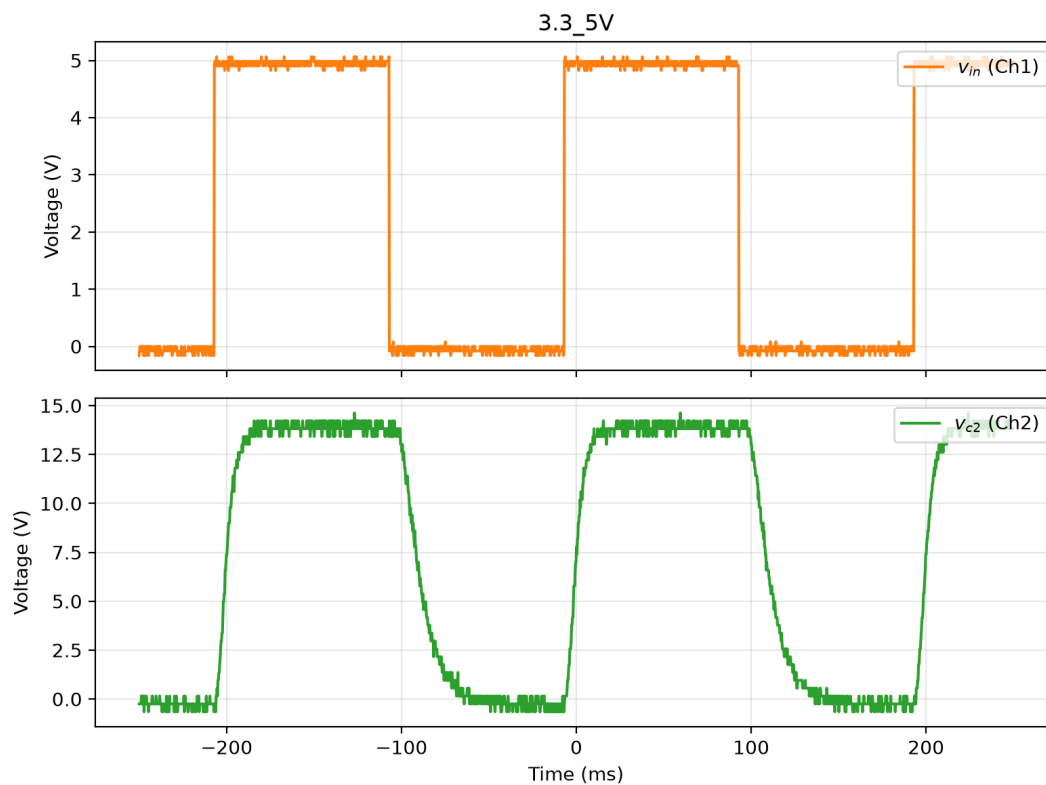


Figure 15: Measured response to a 5 V step (asymmetric rising/falling edges).

The 500 mV step produced a smooth response with similar rise and fall behavior. The output reached about 0.7 V from a 0.5 V input, giving a measured gain of about 1.4. For the 5 V step, the output reached about 7.1 V and the rising and falling edges were no longer as similar. The output also appears to be limited near 7 V, also showing a gain of 1.42.

Task 3.4: Frequency Response Measurements

Table 1: Measured amplitude and time shift of v_{c1} and v_{c2} .

f (Hz)	V_{c1} (pp)	Δt_1 (ms)	V_{c2} (pp)	Δt_2 (ms)
5	0.94	9.5	5.37	14.2
15	0.71	7.9	3.80	12.4
30	0.47	5.8	2.01	9.5
60	0.25	3.5	0.71	6.3
120	0.13	1.9	0.20	3.6

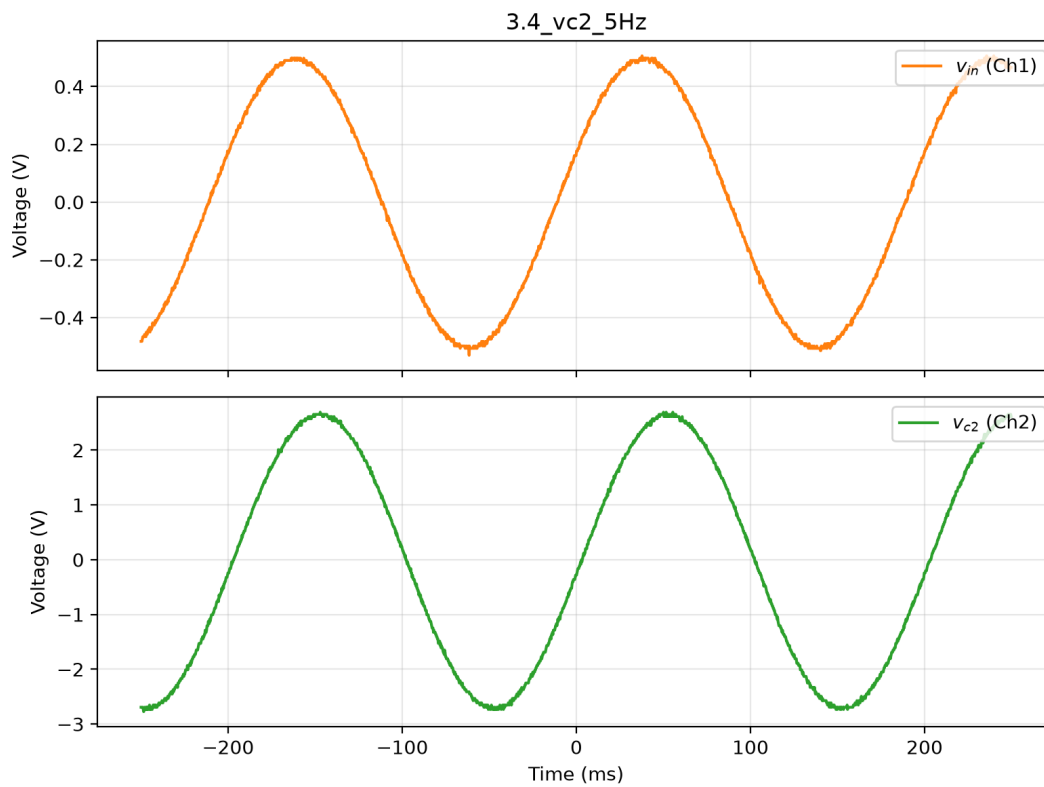


Figure 16: Measured v_{c2} response to a 5 Hz sinusoidal input.

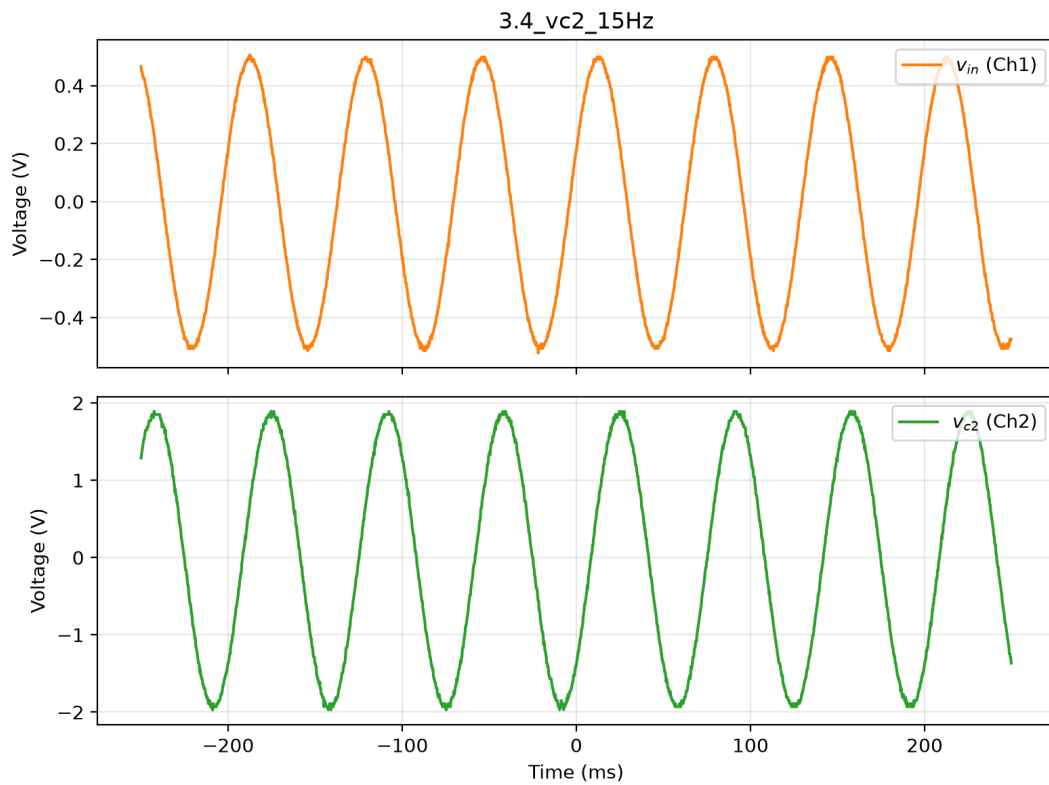


Figure 17: Measured v_{c2} response to a 15 Hz sinusoidal input.

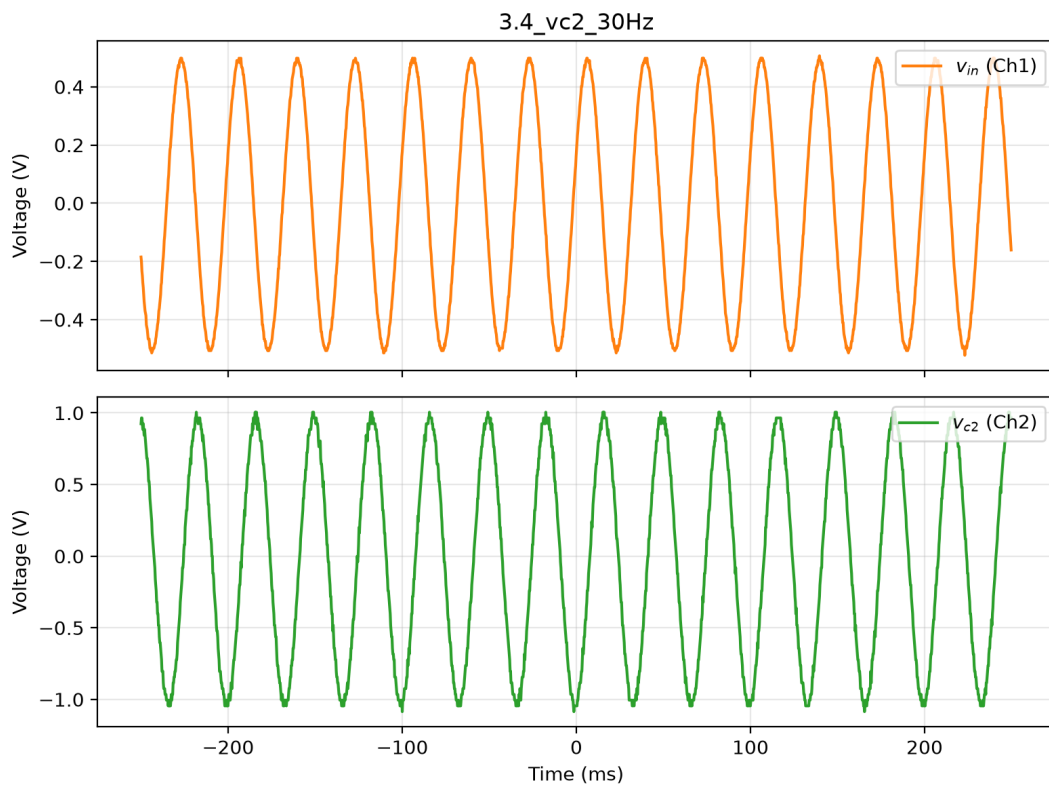


Figure 18: Measured v_{c2} response to a 30 Hz sinusoidal input.

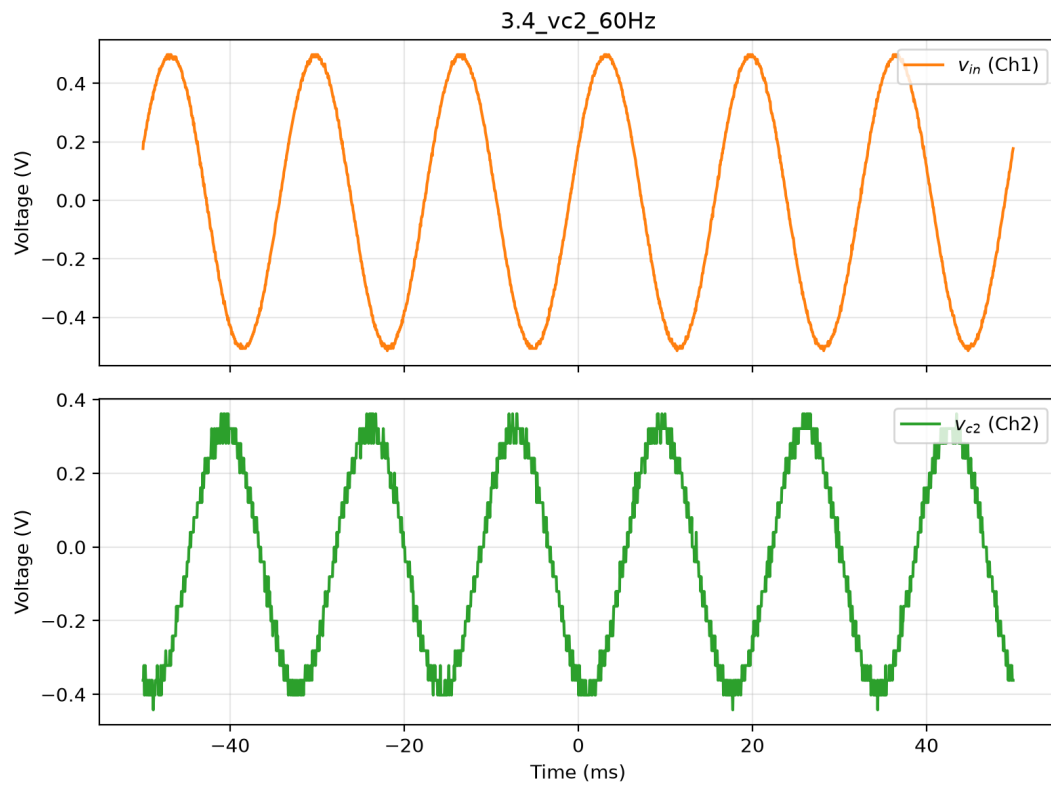


Figure 19: Measured v_{c2} response to a 60 Hz sinusoidal input.

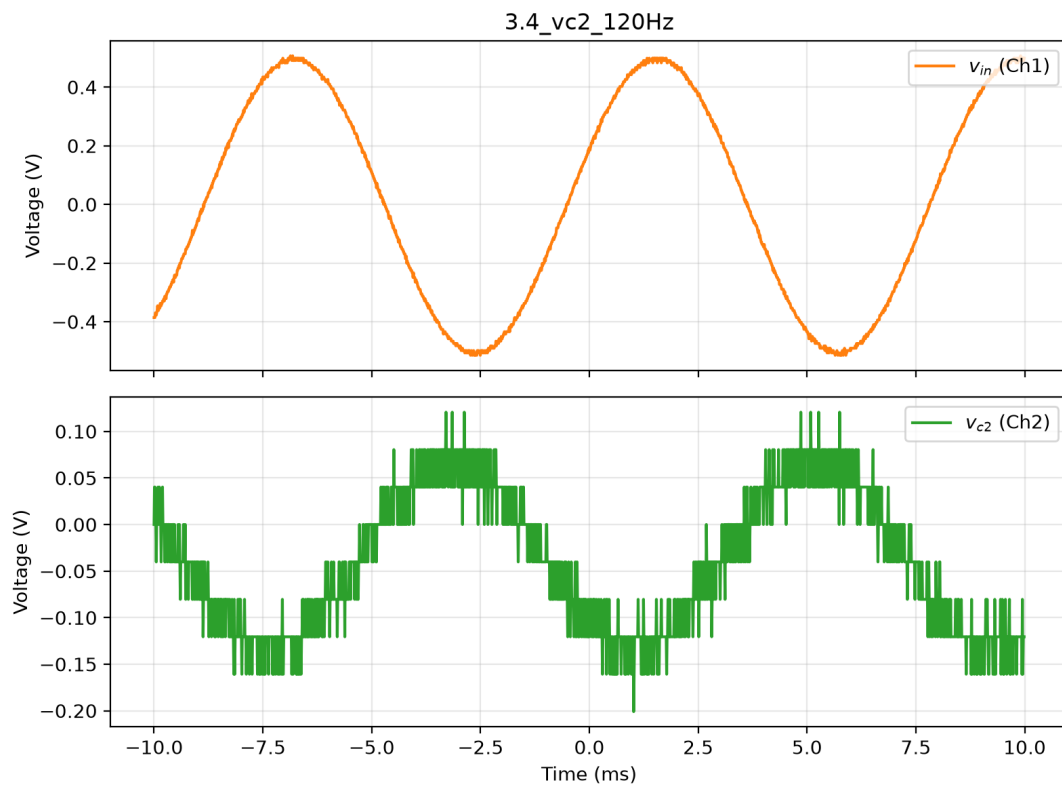


Figure 20: Measured v_{c2} response to a 120 Hz sinusoidal input.

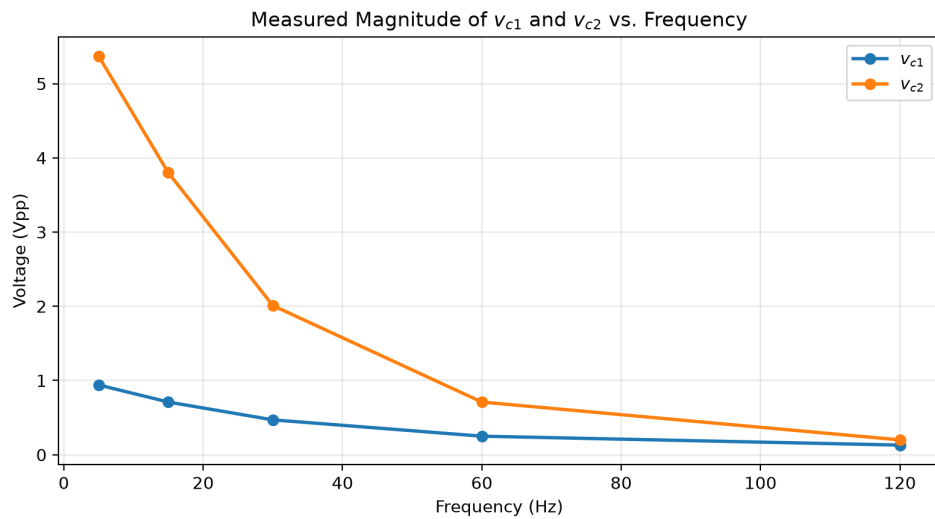


Figure 21: Measured magnitude of v_{c1} and v_{c2} versus frequency.

V_{c2} Has a much larger roll off going from the same frequency it drops from, 5.37 to .2, and V_{c1} goes from .94 to .13V which is far slower roll off at least relative to V_{c1} .

Conclusion

Appendix

Task 3 Code

```

1 import glob, os
2 import pandas as pd
3 import matplotlib
4 matplotlib.use("Agg")
5 import matplotlib.pyplot as plt
6
7 FOLDER = os.path.dirname(os.path.abspath(__file__)) # put this
               script in the CSV folder
8 OUT = os.path.join(FOLDER, "graphs") # separate folder so scope .
               png files never get overwritten
9 os.makedirs(OUT, exist_ok=True)
10
11 for path in sorted(glob.glob(os.path.join(FOLDER, "*.csv"))):
12     name = os.path.splitext(os.path.basename(path))[0]
13
14     # skip the scope header, data starts at the line
15     with open(path, encoding="utf-8-sig") as f:
16         skip = next(i for i, line in enumerate(f) if line.startswith(
17                     "Sample Number"))
18     df = pd.read_csv(path, skiprows=skip, encoding="utf-8-sig").
19         dropna(axis=1, how="all")
20
21     t = df["Time (s)"] * 1e3 # ms
22     ch2 = r"$v_{c1}$" if "vc1" in name else r"$v_{c2}$"
23
24     fig, ax = plt.subplots(figsize=(8, 4.5))
25     ax.plot(t, df.iloc[:, 2], label=r"$v_{in}$ (Ch1)")
26     ax.plot(t, df.iloc[:, 3], label=f"{ch2} (Ch2)")
27     ax.set_xlabel("Time (ms)")
28     ax.set_ylabel("Voltage (V)")
29     ax.set_title(name)
30     ax.grid(True, alpha=0.3)
31     ax.legend(loc="upper right")
32     fig.tight_layout()
33     fig.savefig(os.path.join(OUT, name + ".png"), dpi=200)
34     plt.close(fig)
35     print("saved", name)
36
37 # Task 3.4 frequency response plot
38 frequency = [5, 15, 30, 60, 120]
39 vc1 = [0.94, 0.71, 0.47, 0.25, 0.13]
40 vc2 = [5.37, 3.80, 2.01, 0.71, 0.20]
41
42 fig, ax = plt.subplots(figsize=(8, 4.5))
43
44 ax.plot(frequency, vc1, marker="o", linewidth=2, label=r"$v_{c1}$")
45 ax.plot(frequency, vc2, marker="o", linewidth=2, label=r"$v_{c2}$")
46
47 ax.set_xlabel("Frequency (Hz)")
48 ax.set_ylabel("Voltage (Vpp)")
49 ax.set_title("Measured Magnitude of $v_{c1}$ and $v_{c2}$ vs.
               Frequency")

```

```
49 ax.grid(True, alpha=0.3)
50 ax.legend()
51
52 fig.tight_layout()
53 fig.savefig(os.path.join(OUT, "task3_4_frequency_response.png"), dpi
    =200)
54 plt.close(fig)
```

Listing 3: Graphing raw CSVs and Table data for

```
1 import glob, os
2 import pandas as pd
3 import matplotlib
4 matplotlib.use("Agg")
5 import matplotlib.pyplot as plt
6
7 FOLDER = os.path.dirname(os.path.abspath(__file__)) # put this
              script in the CSV folder
8 OUT = os.path.join(FOLDER, "graphs") # separate folder so scope .
              png files never get overwritten
9 os.makedirs(OUT, exist_ok=True)
10
11 for path in sorted(glob.glob(os.path.join(FOLDER, "*.csv"))):
12     name = os.path.splitext(os.path.basename(path))[0]
13
14     # skip the scope header, data starts at the line
15     with open(path, encoding="utf-8-sig") as f:
16         skip = next(i for i, line in enumerate(f) if line.startswith(
17             "Sample Number"))
18     df = pd.read_csv(path, skiprows=skip, encoding="utf-8-sig").
19         dropna(axis=1, how="all")
20
21     t = df["Time (s)"] * 1e3 # ms
22     ch2 = r"$v_{c1}$" if "vc1" in name else r"$v_{c2}$"
23
24     fig, ax = plt.subplots(figsize=(8, 4.5))
25     ax.plot(t, df.iloc[:, 2], label=r"$v_{in}$ (Ch1)")
26     ax.plot(t, df.iloc[:, 3], label=f"{ch2} (Ch2)")
27     ax.set_xlabel("Time (ms)")
28     ax.set_ylabel("Voltage (V)")
29     ax.set_title(name)
30     ax.grid(True, alpha=0.3)
31     ax.legend(loc="upper right")
32     fig.tight_layout()
33     fig.savefig(os.path.join(OUT, name + ".png"), dpi=200)
34     plt.close(fig)
35     print("saved", name)
```

Listing 4: Normalized Graph